OPERATIONS MANAGEMENT

RESOURCE PLANNING

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Resource Planning

- Resource planning lies at the heart of any organization, cutting across all of its different functional areas.

- It takes sales and operations plans; processes information in the way of time standards, routings, and other information on how services or products are produced; and then plans the input requirements.

- It also can create reports for managers of the firm's major functional areas, such as human resources, purchasing, sales and marketing, and finance and accounting.

- Companies can also gain a competitive edge by using an effective information system to help with their resource planning. Companies must ensure that all of the resources they need to produce finished services or products are available at the right time. If they are not, a firm risks losing business.

- For a manufacturer, this task can mean keeping track of thousands of subassemblies, components, and raw materials as well as key equipment capacities.

- For a service provider, this task can mean keeping track of numerous supplies and carefully scheduling the time and capacity requirements of different employees and types of equipment.

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Enterprise Resource Planning

- An enterprise process is a companywide process that cuts across functional areas, business units, geographic regions, product lines, suppliers, and customers.

- Enterprise resource planning (ERP) systems are large, integrated information systems that support many enterprise processes and data storage needs.

- By integrating the firm's functional areas, ERP systems allow an organization to view its operations as a whole rather than having to try to put together the different information pieces produced by its various functions and divisions.

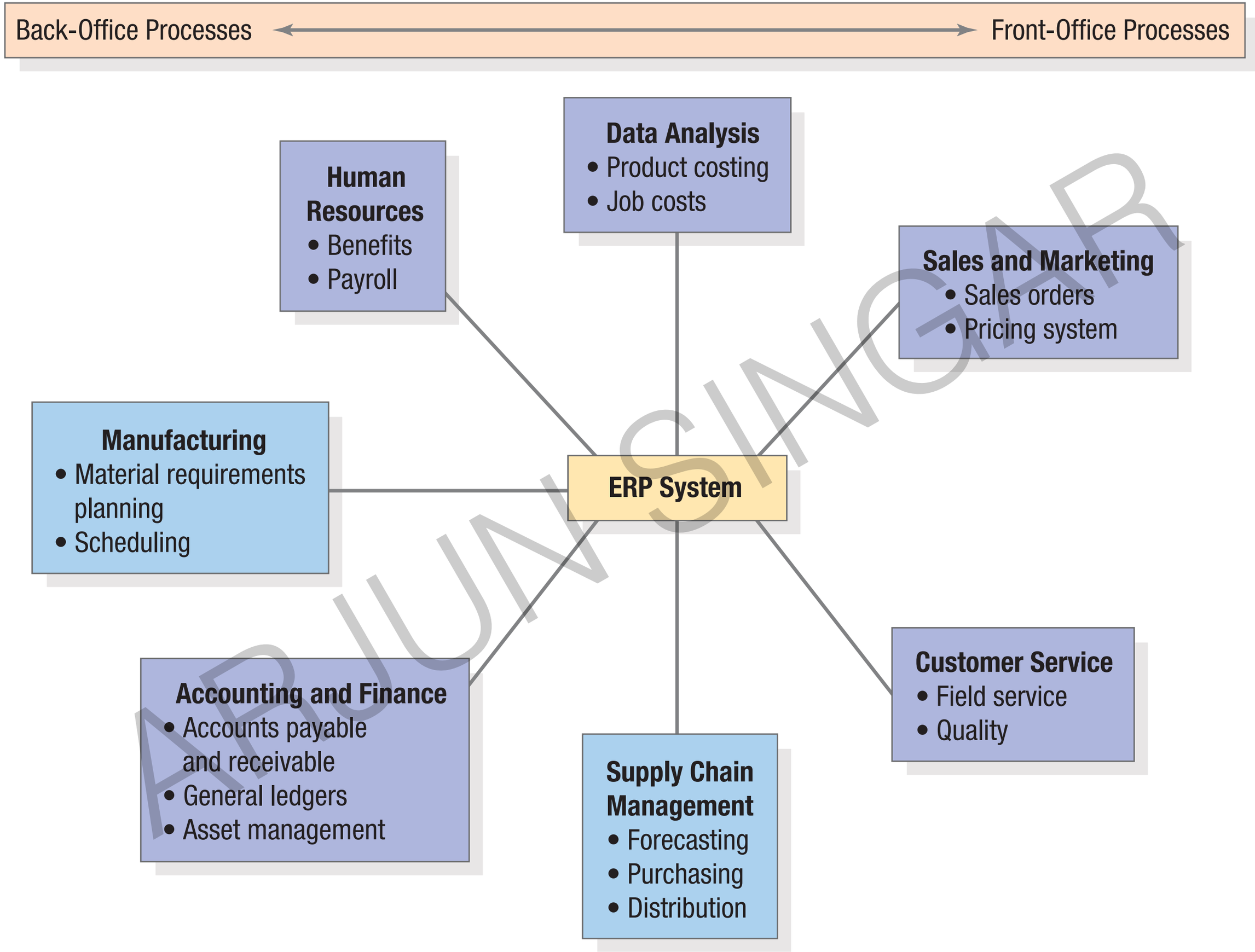
- Today, ERP systems are being used by traditional brick-and-mortar organizations such as manufacturers, restaurants, hospitals, and hotels, as well as by Internet companies that rely extensively on Web connectivity to link their customers and suppliers.

- ERP revolves around a single comprehensive database that can be made available across the entire organization (or enterprise). Having a single database for all of the firm's information makes it much easier for managers to monitor all of the company's products at all locations and at all times.

- The database collects data and feeds them into the various modular applications (or suites) of the software system. As new information is entered as a *transaction* in one application, related information is automatically updated in the other applications, including the firm's financial and accounting databases, its human resource and payroll databases, sales, supplier and customer databases, and so forth.

- In this way, the ERP system streamlines the data flows throughout the organization and supply chain and provides employees with direct access to a wealth of real-time operating information scattered across different functions in the organization.

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Material Requirements Planning

- Material requirements planning **(MRP)** is a computerized information system developed specifically to help manufacturers manage dependent demand inventory and schedule replenishment orders.

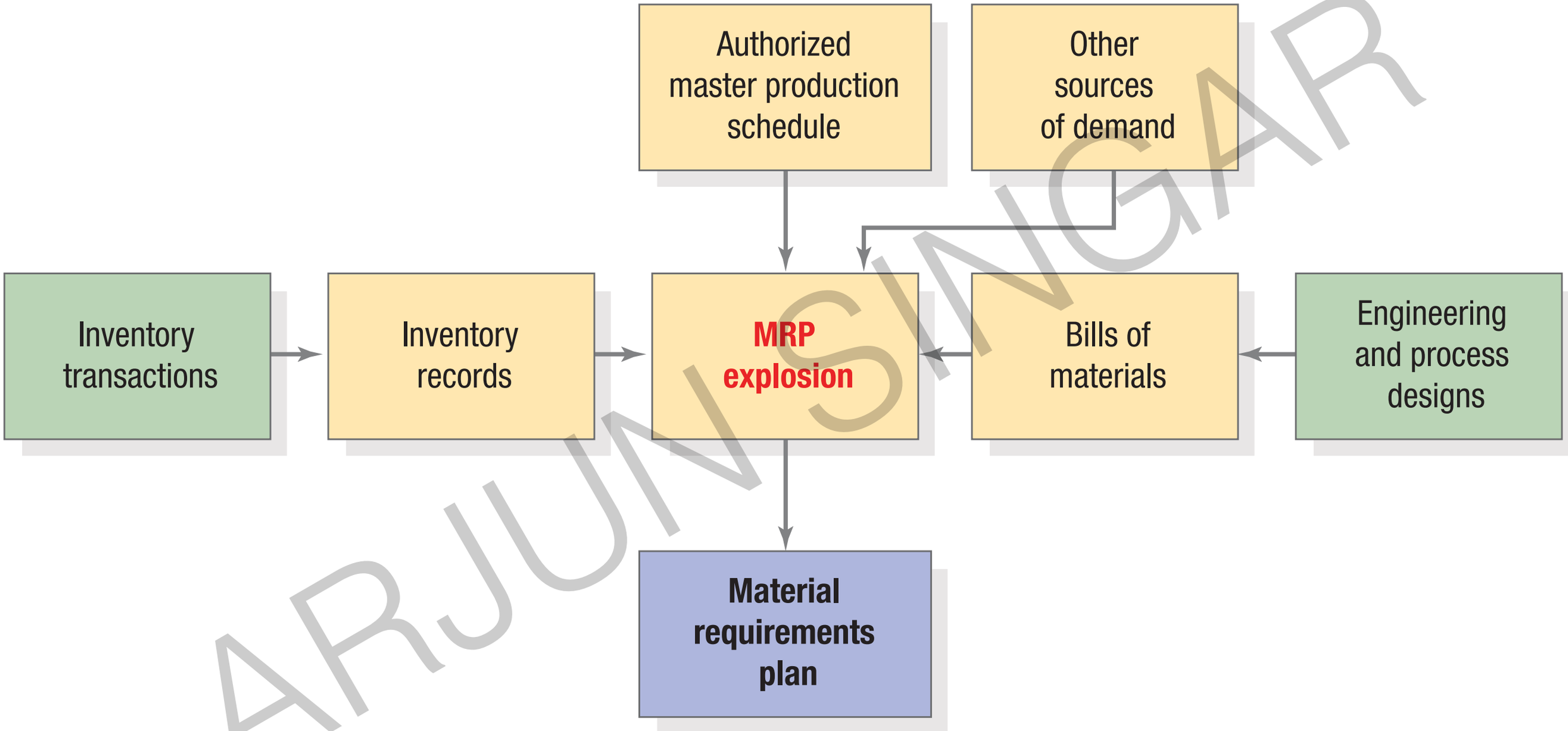
- The **key inputs of an MRP system** are a bill of materials database, a master production schedule, and an inventory record database.

- Using this information, the MRP system identifies the actions planners must take to stay on schedule, such as releasing new production orders, adjusting order quantities, and expediting late orders.

- An MRP system translates the master production schedule and other sources of demand, such as independent demand for replacement parts and maintenance items, into the requirements for all subassemblies, components, and raw materials needed to produce the required parent items.

- This process is called an **MRP explosion** because it converts the requirements of various final products into a material requirements plan that specifies the replenishment schedules of all the subassemblies, components, and raw materials needed to produce final products.

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Dependent Demand

- To illustrate the concept of dependent demand, let us consider a Huffy bicycle produced for retail outlets. Demand for a final product, such as a bicycle, is called *independent demand* because it is influenced only by market conditions.

- In contrast, the demand for spokes going into the bicycle "depends" on the production planned for its wheels. Huffy also keeps many other items in inventory —handlebars, pedals, frames, and wheel rims—used to make completed bicycles. Each of these items has a **dependent demand** because the quantity required varies with the production plans for other items held in the firm's inventory—finished bikes, in this case.

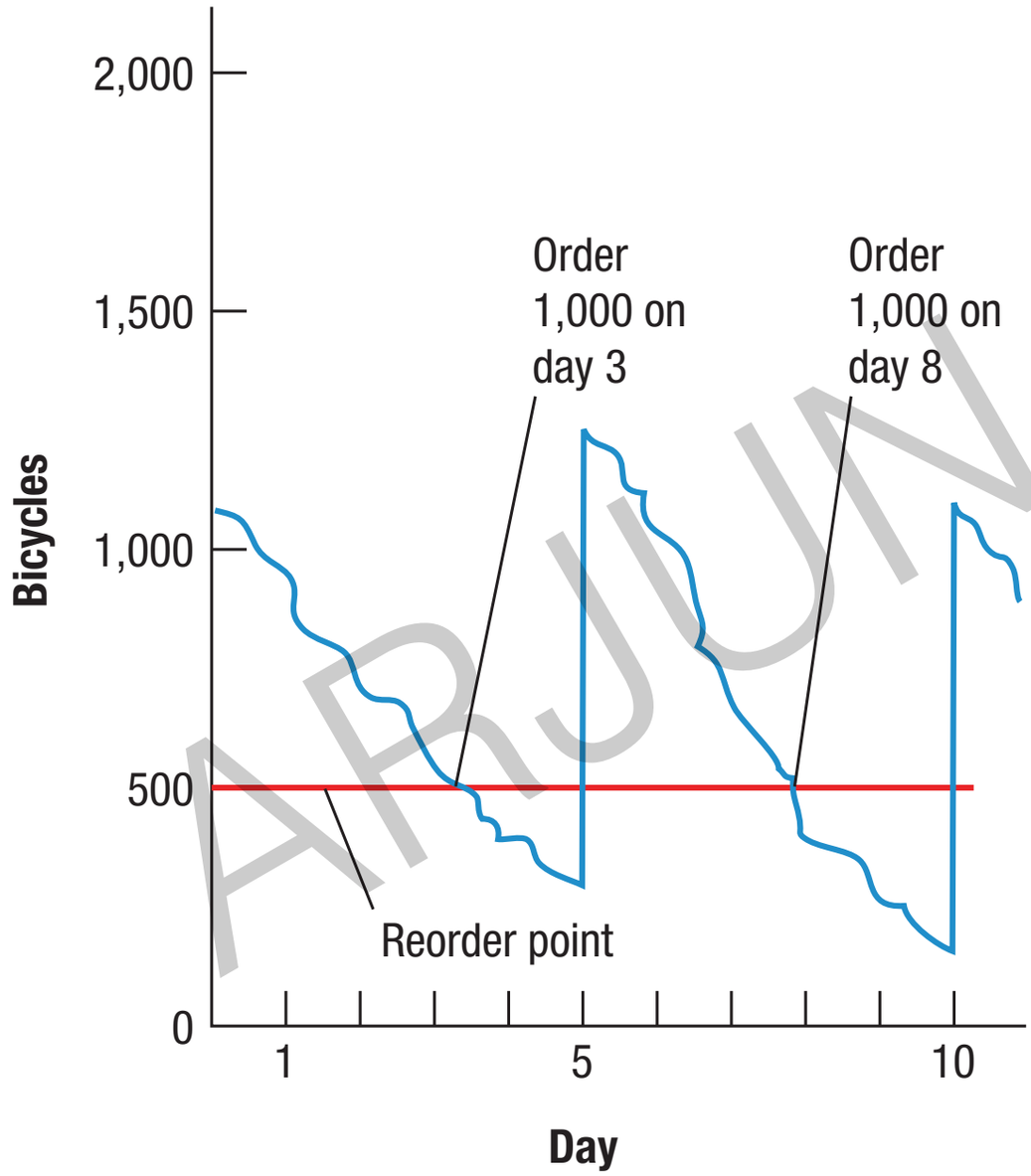
- For example, the demand for frames, pedals, and wheel rims is *dependent* on the production of completed bicycles. Operations can calculate the demand for dependent demand items once the bicycle production levels are laid out in the sales and operations plan.

- For example, every bicycle needs two wheel rims, so 1,000 completed bicycles need 1,000(2) = 2,000 rims.

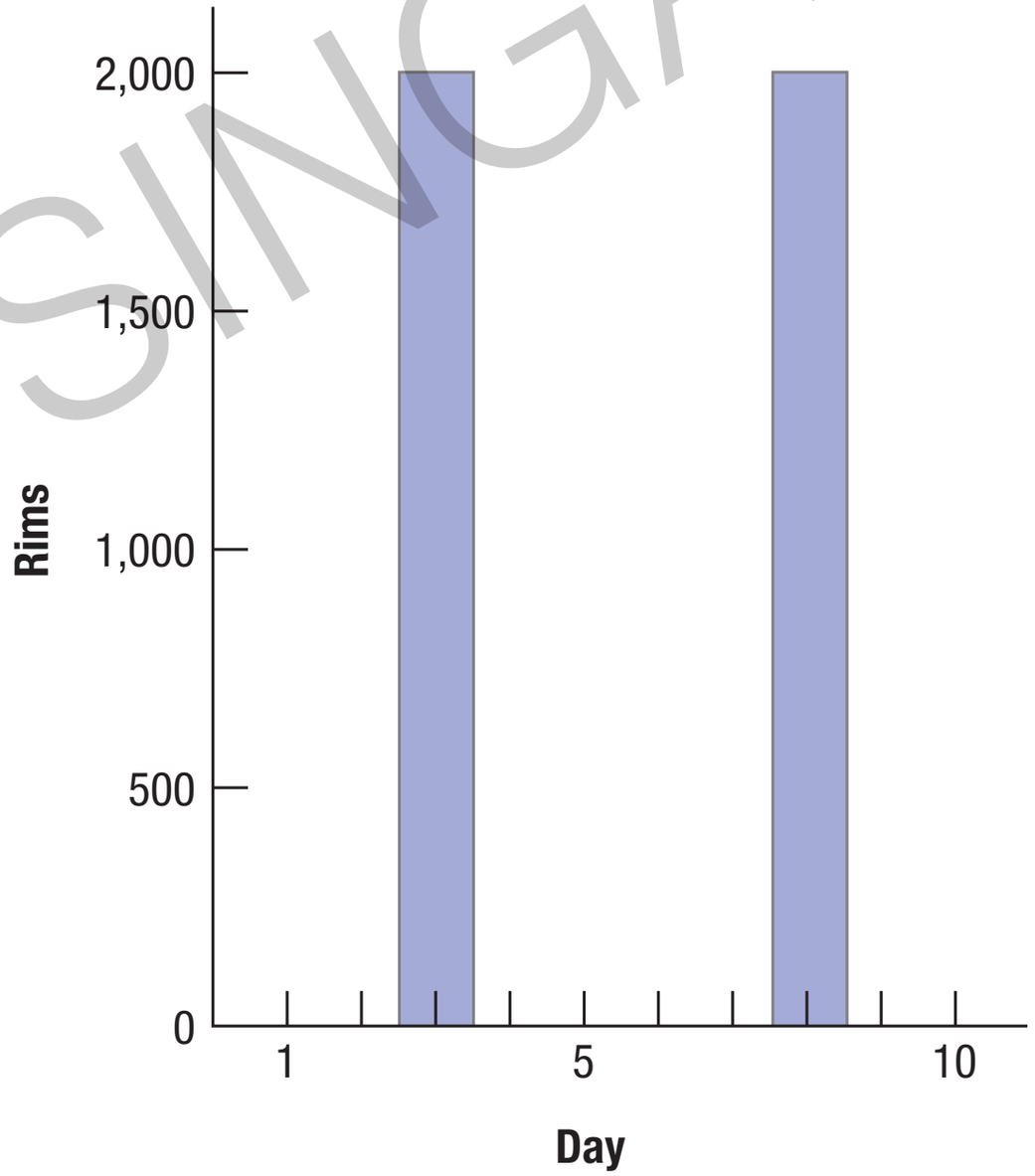
- The bicycle, or any other product that is manufactured from one or more components, is called a **parent**. The wheel rim is an example of a **component** — an item that goes through one or more operations to be transformed into or become part of one or more parents. A wheel rim, for example, will have several different parents if the rim is used to make more than one style of bicycle.

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- Suppose that every time inventory falls to 500 units (a reorder point), an order for 1,000 more bicycles is placed. The assembly supervisor then authorizes the withdrawal of 2,000 rims from inventory, along with other components for the finished product. So, even though customer demand for the finished bicycle is continuous and reasonably uniform, the production demand for wheel rims is "lumpy"; that is, it occurs sporadically, usually in relatively large quantities.



(a) Parent inventory



(b) Component demand

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Bill of Materials

- The replenishment schedule for a component is determined from the production schedules of its parents. Hence, the system needs accurate information on parent – component relationships.

- The first input into a MRP is BOM. A **bill of materials (BOM)** is a record of all the components of an item, the parent–component relationships, and the usage quantities derived from engineering and process designs.

- In the following Figure, the BOM of a simple ladder-back chair shows that the chair is made from a ladder-back subassembly, a seat subassembly, front legs, and leg supports. In turn, the ladder-back subassembly is made from back legs and back slats, and the seat subassembly is made from a seat frame and a seat cushion. Finally, the seat frame is made from seat-frame boards.

- The BOM also specifies the **usage quantity**, or the number of units of a component that are needed to make one unit of its immediate parent.

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- Four terms frequently used to describe inventory items are *end items*, *intermediate items*, *sub-assemblies*, and *purchased items*.

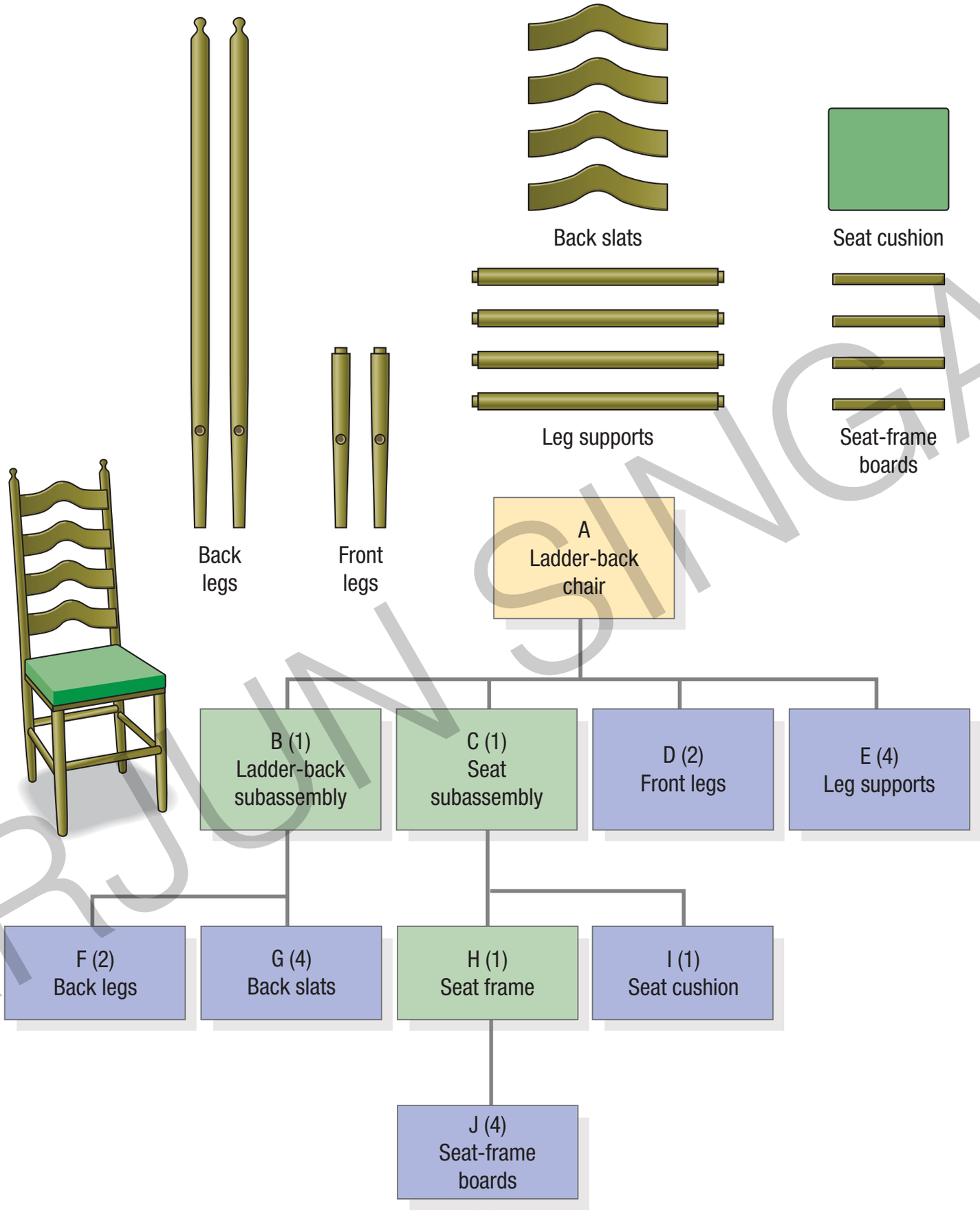
- An **end item** typically is the final product sold to the customer; it is a parent but not a component. Item A, the completed ladder-back chair, is an end item.

- An **intermediate item**, such as item B, C, or H, has at least one parent and at least one component. Some products have several levels of intermediate items; the parent of one intermediate item can also be an intermediate item.

- A **subassembly** is an inter-mediate item that is assembled (as opposed to being transformed by other means) from more than one component. Items B and C are subassemblies.

- A **purchased item** has no components because it comes from a supplier, but it has one or more parents. Examples are items D, E, F, G, I, and J.

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iPad total product costs $270		iPad 2 total product costs $267
	LCD display – 9.7-inch multi-touch screen **$59**	LCD display – 9.7-inch multi-touch screen **$50**
	Camera – None **$0**	Camera – Front & Rear **$4.50**
	Memory – Samsung **$47**	Memory – Samsung/Toshiba **$30**
	Apple Processor – Apple A4 **$17**	Apple Processor – Apple A5 **$25**
	Radio Components Infineon/Broadcom **$26**	Radio Components Infineon/Broadcom or Qualcomm/Broadcom **$25**
	Sensors – STMicroelectronics **$1.00**	Sensors – STMicroelectronics **$2.50**
	Battery **$23**	Battery **$20 – $25**
		

Bill of Materials for Apple's iPad and iPad2.

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Master Production Scheduling

- The second input into a material requirements plan is the master production schedule **(MPS)**, which details how many end items will be produced within specified periods of time. It breaks the sales and operations plan into specific product schedules.

- The following Figure shows how a sales and operations plan for a family of chairs breaks down into the weekly MPS for each specific chair type (the time period can be hours, days, weeks, or months). The chair example demonstrates the following aspects of master scheduling:

1. The sums of the quantities in the MPS must equal those in the sales and operations plan.

2. The production quantities must be allocated efficiently over time. The specific mix of chair types—the number of each type as a percent of the total family's quantity—is based on historic demand and on marketing and promotional considerations.

3. Capacity limitations and bottlenecks, such as machine or labor capacity, storage space, or working capital, may determine the timing and size of MPS quantities.

- The process of developing a master production schedule includes **(1)** calculating the projected on-hand inventory and **(2)** determining the timing and size of the production quantities of specific products.

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- Operations must first create a prospective MPS to test whether it meets the schedule with the resources (e.g., machine capacities, workforce, overtime, and subcontractors) provided for in the sales and operations plan.

- Operations then revises the MPS until a schedule that satisfies all of the resource limitations is developed or until it is determined that no feasible schedule can be developed.

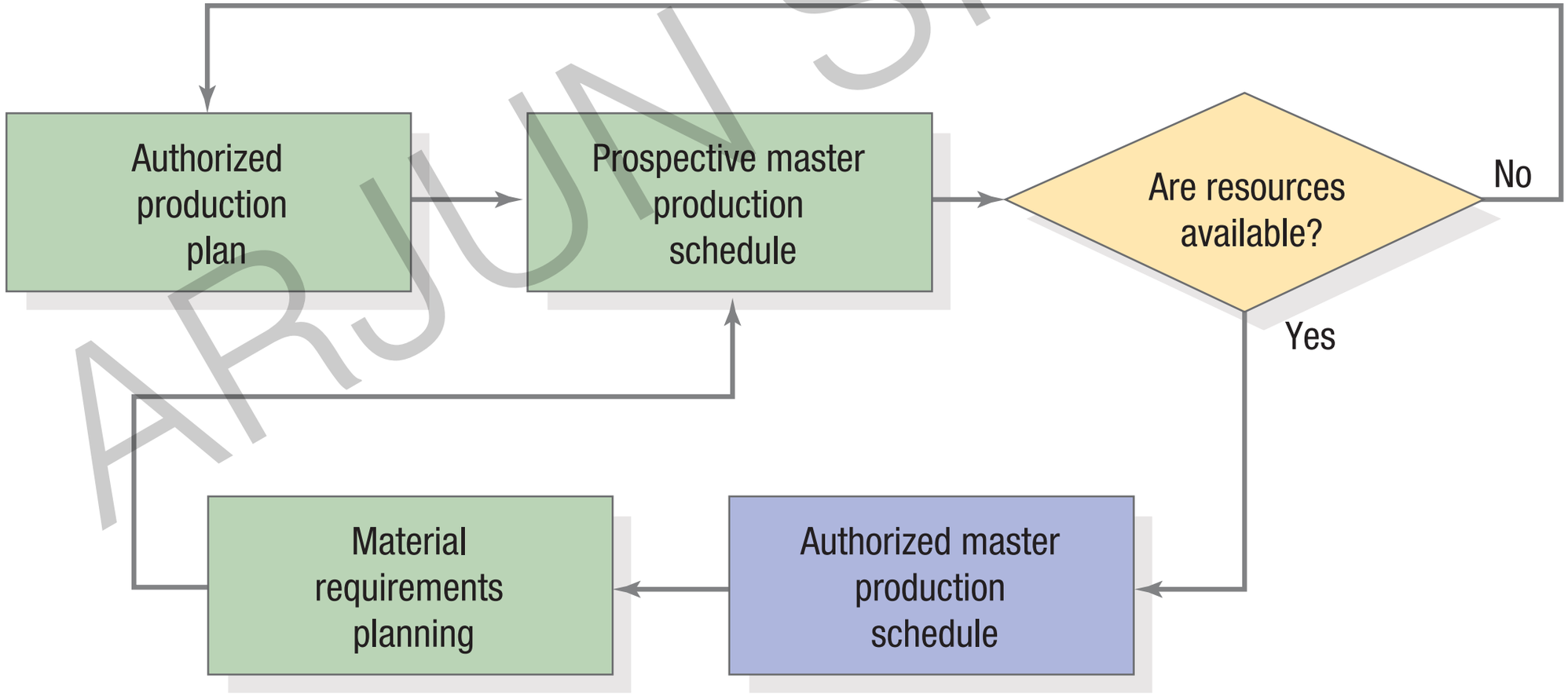
- In the latter event, the production plan must be revised to adjust production requirements or increase authorized resources.

- Once a feasible prospective MPS has been accepted by the firm's managers, operations uses the authorized MPS as input to material requirements planning.

- Operations can then determine specific schedules for component production and assembly.

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	April				May			
	1	2	3	4	5	6	7	8
Ladder-back chair	150					150		
Kitchen chair				120			120	
Desk chair		200	200		200			200
Sales and operations plan for chair family	670				670			



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Developing a Master Production Schedule

- ***Step 1.*** *Calculate Projected On-Hand Inventories:* The first step is to calculate the projected on-hand inventory, which is an estimate of the amount of inventory available each week after demand has been satisfied.

- In some weeks, no MPS quantity for a product may be needed because sufficient inventory already exists.

- For the projected requirements for this week, the scheduler uses whichever is larger — the forecast or the customer orders booked — recognizing that the forecast is subject to error.

- If actual booked orders exceed the forecast, the projection will be more accurate if the scheduler uses the booked orders because booked orders are a known quantity. Conversely, if the forecast exceeds booked orders for a week, the forecast will provide a better estimate of the requirements needed for that week because some orders are yet to come in.

$$\begin{pmatrix}\text{Projected on-hand}\\ \text{inventory at end}\\ \text{of this week}\end{pmatrix} = \begin{pmatrix}\text{On-hand}\\ \text{inventory at}\\ \text{end of last week}\end{pmatrix} + \begin{pmatrix}\text{MPS quantity}\\ \text{due at start}\\ \text{of this week}\end{pmatrix} - \begin{pmatrix}\text{Projected}\\ \text{requirements}\\ \text{this week}\end{pmatrix}$$

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Item: Ladder-back chair

Quantity on Hand: 55	April 1	April 2	
Forecast	30	30	
Customer orders (booked)	38	27	
Projected on-hand inventory	17	–13	
MPS quantity	0	0	
MPS start			

Explanation:
Forecast is less than booked orders in week 1; projected on-hand inventory balance
= 55 + 0 – 38 = 17.

Explanation:
Forecast exceeds booked orders in week 2; projected on-hand inventory balance
= 17 + 0 – 30 = –13. The shortage signals a need to schedule an MPS quantity for completion in week 2.

FIGURE 16.7
Master Production Schedule for Weeks 1 and 2

$$\text{Inventory} = \begin{pmatrix}\text{55 chairs}\\ \text{currently}\\ \text{in stock}\end{pmatrix} + \begin{pmatrix}\text{MPS quantity}\\ \text{(0 for week 1}\end{pmatrix} - \begin{pmatrix}\text{38 chairs already}\\ \text{promised for}\\ \text{delivery in week 1}\end{pmatrix} = 17 \text{ chairs}$$

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- ***Step 2.*** *Determine the Timing and Size of MPS Quantities:* The goal of determining the timing and size of MPS quantities is to maintain a nonnegative projected on-hand inventory balance.

- As shortages in inventory are detected, MPS quantities should be scheduled to cover them.

- The first MPS quantity should be scheduled for the week when the projected on-hand inventory reflects a shortage, such as week 2 in the following figure.

- The scheduler adds the MPS quantity to the projected on-hand inventory and searches for the next period when a shortage occurs. This shortage signals a need for a second MPS quantity, and so on.

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Item: Ladder-back chair

Order Policy: 150 units
Lead Time: 1 week

Quantity on Hand: 55	**April**				**May**			
	1	2	3	4	5	6	7	8
Forecast	30	30	30	30	35	35	35	35
Customer orders (booked)	38	27	24	8	0	0	0	0
Projected on-hand inventory	17	137	107	77	42	7	122	87
MPS quantity		150					150	
MPS start	150					150		

Explanation:
The time needed to assemble 150 chairs is 1 week. The assembly department must start assembling chairs in week 1 to have them ready by week 2.

Explanation:
On-hand inventory balance = 17 + 150 – 30 = 137. The MPS quantity is needed to avoid a shortage of 30 – 17 = 13 chairs in week 2.

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- **Available-to-Promise Quantities:** In addition to providing manufacturing with the timing and size of production quantities, the MPS provides marketing with information useful for negotiating delivery dates with customers.

- The quantity of end items that marketing can promise to deliver on specified dates is called **available-to-promise (ATP) inventory**. **It is the difference between the customer orders already booked (until the next MPS receipt) and the quantity that operations is planning to produce (MPS Quantity)**.

- As new customer orders are accepted, the ATP inventory is reduced to reflect the commitment of the firm to ship those quantities, but the actual inventory stays unchanged until the order is removed from inventory and shipped to the customer.

- **An available-to-promise inventory is associated with each MPS quantity because the MPS quantity specifies the timing and size of new stock that can be earmarked to meet future bookings**.

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Item: Ladder-back chair

Order Policy: 150 units
Lead Time: 1 week

Quantity on Hand: 55	**April**				**May**			
	1	2	3	4	5	6	7	8
Forecast	30	30	30	30	35	35	35	35
Customer orders (booked)	38	27	24	8	0	0	0	0
Projected on-hand inventory	17	137	107	77	42	7	122	87
MPS quantity		150					150	
MPS start	150					150		
Available-to-promise (ATP) inventory	17	91					150	

Explanation:
The total of customer orders booked until the next MPS receipt is 38 units. The ATP = 55 (on-hand) + 0 (MPS quantity) – 38 = 17.

Explanation:
The total of customer orders booked until the next MPS receipt is 27 + 24 + 8 = 59 units. The ATP = 150 (MPS quantity) – 59 = 91 units.

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Inventory Record

- Inventory records are a third major input to MRP, and inventory transactions are the basic building blocks of up-to-date records.

- These transactions include releasing new orders, receiving scheduled receipts, adjusting due dates for scheduled receipts, withdrawing inventory, canceling orders, correcting inventory errors, rejecting shipments, and verifying scrap losses and stock returns.

- Recording the transactions accurately is essential if the firm's on-hand inventory balances are to be correct and its MRP system is to operate effectively.

- The **inventory record** divides the future into time periods called *time buckets*. In our discussion, we use weekly time buckets for consistency with our MPS example, although other time periods could as easily be used.

- The inventory record shows an item's lot-size policy, lead time, and various time-phased data. The **purpose of the inventory record** is to keep track of inventory levels and component replenishment needs.

- The time-phased information contained in the inventory record consists of (1) *gross requirements*, (2) *scheduled receipts*, (3) *projected on-hand inventory*, (4) *planned receipts*, and (5) *planned order releases*.

- We illustrate the discussion of inventory records with the seat subassembly, item C that was shown in the BOM Figure. Suppose that it is used in two products: a ladder-back chair and a kitchen chair.

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- The **gross requirements** are the total demand derived from *all* parent production plans. They also include demand not otherwise accounted for, such as demand for replacement parts for units already sold.

- The **scheduled receipts** (sometimes called open orders) are orders that have been placed but not yet completed. For a purchased item, the scheduled receipt could be in one of several stages: being processed by a supplier, being transported to the purchaser, or being inspected by the purchaser's receiving department. If the firm is making the item in-house, the order could be on the shop floor being processed, waiting for components, waiting for a machine to become available, or waiting to be moved to its next operation.

- The **projected on-hand inventory** is an estimate of the amount of inventory available each week after gross requirements have been satisfied. The beginning inventory, shown as the first entry (37) in the following figure, indicates the on-hand inventory available at the time the record was computed. As with scheduled receipts, entries are made for each actual withdrawal and receipt to update the MRP database. Then, when the MRP system produces the revised record, the correct inventory will appear.

$$\begin{pmatrix}\text{Projected on-hand}\\ \text{inventory balance}\\ \text{at end of week } t\end{pmatrix} = \begin{pmatrix}\text{Inventory on}\\ \text{hand at end of}\\ \text{week } t-1\end{pmatrix} + \begin{pmatrix}\text{Scheduled}\\ \text{or planned}\\ \text{reeipts in}\\ \text{week } t\end{pmatrix} - \begin{pmatrix}\text{Gross}\\ \text{requirements}\\ \text{in week } t\end{pmatrix}$$

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Item: C
Description: Seat subassembly

Lot Size: 230 units
Lead Time: 2 weeks

	Week							
	1	2	3	4	5	6	7	8
Gross requirements	150	0	0	120	0	150	120	0
Scheduled receipts	230	0	0	0	0	0	0	0
Projected on-hand inventory 37	117	117	117	–3	–3	–153	–273	–273
Planned receipts								
Planned order releases								

Explanation:
Gross requirements are the total demand for the two chairs. Projected on-hand inventory in week 1 is 37 + 230 – 150 = 117 units.

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- The projected on-hand calculation includes the consideration of **planned receipts**, which are orders not yet released to the shop or the supplier. Planned receipts should not be confused with scheduled receipts. Planned receipts are still at the planning stage and can still change from one week to the next, whereas scheduled receipts are actual orders that are being acted upon by the shop or supplier. Planning for the receipt of new orders will keep the projected on-hand balance from dropping below zero.

- A **planned order release** indicates when an order for a specified quantity of an item is to be issued. We must place the planned order release quantity in the proper time bucket. To do so, we must assume that all inventory flows—scheduled receipts, planned receipts, and gross requirements—occur at the same point of time in a time period. Some firms assume that all flows occur at the beginning of a time period; other firms assume that they occur at the end of a time period or at the middle of the time period. Regardless of when the flows are assumed to occur, we find the release date by subtracting the lead time from the receipt date.

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Item: C
Description: Seat subassembly

Lot Size: 230 units
Lead Time: 2 weeks

	Week							
	1	2	3	4	5	6	7	8
Gross requirements	150	0	0	120	0	150	120	0
Scheduled receipts	230	0	0	0	0	0	0	0
Projected on-hand inventory 37	117	117	117	227	227	77	187	187
Planned receipts				230			230	
Planned order releases		230			230			

Explanation:
Without a planned receipt in week 4, a shortage of 3 units will occur: 117 + 0 + 0 – 120 = –3 units. Adding the planned receipt brings the balance to 117 + 0 + **230** – 120 = 227 units. Offsetting for a 2-week lead time puts the corresponding planned order release back to week 2.

Explanation:
The first planned receipt lasts until week 7, when projected inventory would drop to 77 + 0 + 0 – 120 = –43 units. Adding the second planned receipt brings the balance to 77 + 0 + **230** – 120 = 187 units.
The corresponding planned order release is for week 5 (or week 7 minus 2 weeks).

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Planning Factors

- The planning factors in a MRP inventory record play an important role in the overall performance of the MRP system. By manipulating these factors, managers can fine-tune inventory operations. In this section, we discuss **planning lead time, lot-sizing rules, and safety stock**.

- **Planning lead time** is an estimate of the time between placing an order for an item and receiving the item in inventory. Accuracy is important in planning lead time. If an item arrives in inventory sooner than needed, inventory holding costs increase. If an item arrives too late, stockouts, excessive expediting costs, or both may occur.

- For purchased items, the planning lead time is the time allowed for receiving a shipment from the supplier after the order has been sent, including the normal time to place the order. Often, the purchasing contract stipulates the delivery date. For items manufactured in-house, a rough-cut estimate of the planning lead time can be obtained by keeping track of the actual lead times for recent orders and computing an average. A more extensive estimating process consists of breaking down each of the following factors:

(i) Setup time (ii) Processing time (iii) Materials handling time between operations (iv) Waiting time

- Each of these times must be estimated for every operation along the item's route. Estimating setup, processing, and materials handling times can be relatively easy, but estimating the waiting time for materials handling equipment or for a workstation to perform a particular operation can be more difficult. Therefore, being able to accurately estimate the waiting time is especially important when it comes to estimating the planning lead time.

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- A **lot-sizing rule** determines the timing and size of order quantities. A lot-sizing rule must be assigned to each item before planned receipts and planned order releases can be computed. The choice of lot-sizing rules is important because they determine the number of setups required and the inventory holding costs for each item. We present three lot-sizing rules: **(1)** fixed order quantity, **(2)** periodic order quantity, and **(3)** lot-for-lot.

1. The **fixed order quantity (FOQ)** rule maintains the same order quantity each time an order is issued.

2. The **periodic order quantity (POQ)** rule allows a different order quantity for each order issued but issues the order for predetermined time intervals. The order quantity equals the amount of the item needed during the predetermined time between orders and must be large enough to prevent shortages. This amount exactly covers *P* weeks' worth of gross requirements. That is, the projected on-hand inventory should equal zero at the end of the *P*th week.

3. A special case of the POQ rule is the **lot-for-lot (L4L) rule**, under which the lot size ordered covers the gross requirements of a single week. Thus, $P = 1$, and the goal is to minimize inventory levels. This rule ensures that the planned order is just large enough to prevent a shortage in the single week it covers. The projected on-hand inventory combined with the new order will equal zero at the end of week *t*. Following the first planned order, an additional planned order will be used to match each subsequent gross requirement.

$$\begin{pmatrix}\text{POQ lot size}\\ \text{to arrive in}\\ \text{week } t\end{pmatrix} = \begin{pmatrix}\text{Total gross requirements}\\ \text{for } P \text{ weeks, including}\\ \text{week } t\end{pmatrix} - \begin{pmatrix}\text{Projected on-hand}\\ \text{inventory balance at}\\ \text{end of week } t-1\end{pmatrix}$$

$$\begin{pmatrix}\text{L4L lot size}\\ \text{to arrive in}\\ \text{week } t\end{pmatrix} = \begin{pmatrix}\text{Gross requirements}\\ \text{for week } t\end{pmatrix} - \begin{pmatrix}\text{Projected on-hand}\\ \text{inventory balance at}\\ \text{end of week } t-1\end{pmatrix}$$

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- An important managerial decision is the quantity of **safety stock** to carry. It is more complex for dependent demand items than for independent demand items.

- Safety stock for dependent demand items with lumpy demand (gross requirements) is helpful only when future gross requirements, the timing or size of scheduled receipts, and the amount of scrap that will be produced are uncertain. As these uncertainties are resolved, safety stock should be reduced and ultimately eliminated.

- The usual policy is to use safety stock for end items and purchased items to protect against fluctuating customer orders and unreliable suppliers of components but to avoid using it as much as possible for intermediate items.

- Safety stocks can be incorporated in the MRP logic by using the following rule: Schedule a planned receipt whenever the projected on-hand inventory balance drops below the desired safety stock level (rather than zero, as before). The objective is to keep a minimum level of planned inventories equal to the safety stock quantity.

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Outputs from MRP

- The MRP systems provide many reports, schedules, and notices to help planners control dependent demand inventories.

- MRP translates, or *explodes*, the MPS and other sources of demand into the requirements needed for all of the subassemblies, components, and raw materials the firm needs to produce parent items. This process generates the material requirements plan for each component item.

- An item's gross requirements are derived from three sources:

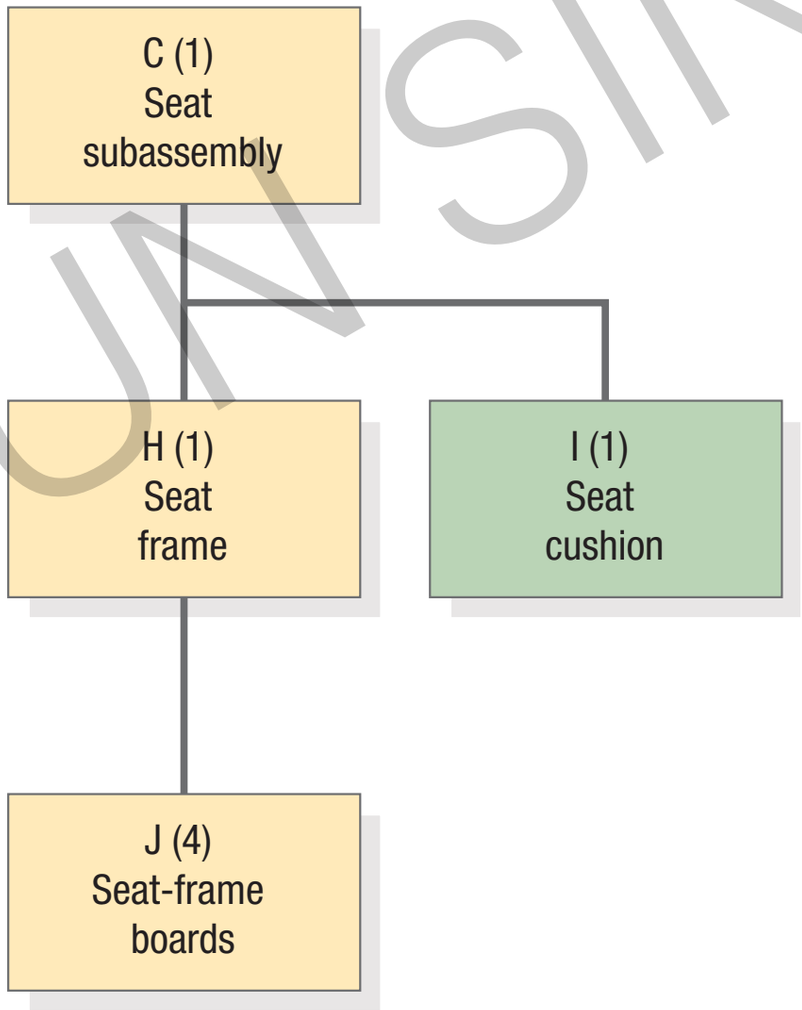
1. The MPS for immediate parents that are end items
2. The planned order releases (*not* the gross requirements, scheduled receipts, or planned receipts) for immediate parents below the MPS level
3. Any other requirements not originating in the MPS, such as the demand for replacement parts

- The key to the explosion process is to determine the proper timing and size of the gross requirements for each component. After we make those determinations, we can derive the planned order release schedule for each component.

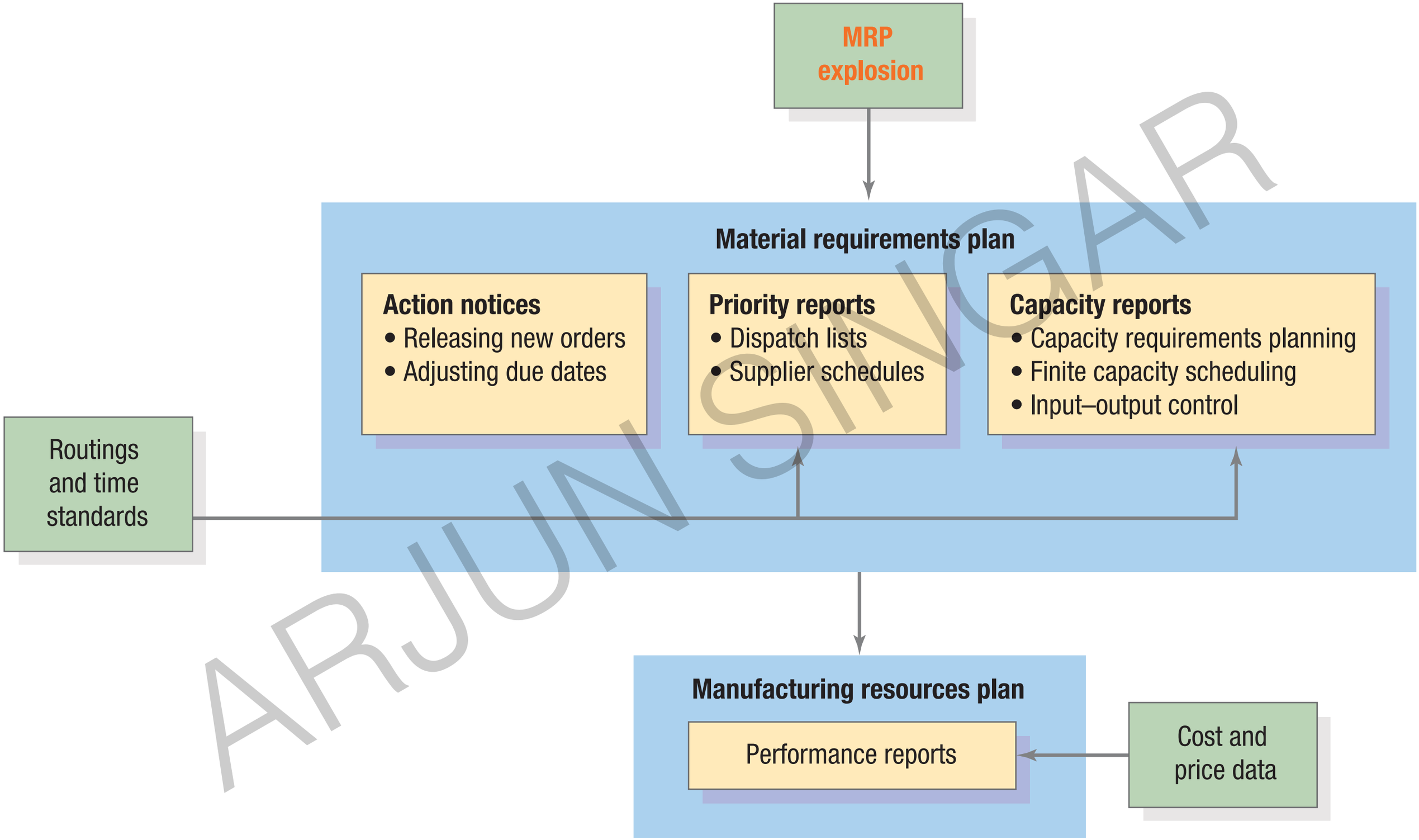
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- Consider the seat subassembly and its inventory record. The seat subassembly requires a seat cushion and a seat frame, which in turn needs four seat-frame boards. Its BOM is shown in the following figure. How many seat cushions should we order from the supplier? How many seat frames should we produce to support the seat subassembly schedule? How many seat-frame boards do we need to make? The answers to these questions depend on the existing inventories of these items and the replenishment orders already in progress.

- MRP can help answer these questions through the explosion process.



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MRP Records for the Seat Subassembly and its Components

Item: Seat subassembly
Lot size: 230 units

Lead time: 2 weeks	**Week** 1	2	3	4	5	6	7	8
Gross requirements	150	0	0	120	0	150	120	0
Scheduled receipts	230	0	0	0	0	0	0	0
Projected inventory 37	117	117	117	227	227	77	187	187
Planned receipts				230			230	
Planned order releases		230			230			

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Usage quantity: 1

Usage quantity: 1

Item: Seat frames
Lot size: 300 units

Lead time: 1 week		**Week**							
		1	2	3	4	5	6	7	8
Gross requirements		0	230	0	0	230	0	0	0
Scheduled receipts		0	300	0	0	0	0	0	0
Projected inventory	40	40	110	110	110	180	180	180	180
Planned receipts						300			
Planned order releases					300				

Item: Seat cushion
Lot size: L4L

Lead time: 1 week		**Week**							
		1	2	3	4	5	6	7	8
Gross requirements		0	230	0	0	230	0	0	0
Scheduled receipts		0	0	0	0	0	0	0	0
Projected inventory	0	0	0	0	0	0	0	0	0
Planned receipts			230			230			
Planned order releases		230			230				

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Usage quantity: 4

Item: Seat-frame boards
Lot size: 1,500 units

Lead time: 1 week	**Week** 1	2	3	4	5	6	7	8
Gross requirements	0	0	0	1,200	0	0	0	0
Scheduled receipts	0	0	0	0	0	0	0	0
Projected inventory 200	200	200	200	500	500	500	500	500
Planned receipts				1,500				
Planned order releases			1,500					

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MRP, Core Processes, and Supply Chain Linkages

- Among the four core processes of an organization that link activities within and across firms in a supply chain, the MRP system interacts with all of them either through its inputs or its outputs. It all begins with customer orders, which consist of orders for end items as well as replacement parts.

- MRP and resource planning typically reside inside the order fulfillment process. Master Production Scheduling is an integral part of MRP. As shown schematically in the following figure, the MPS drives the feedback between the order fulfillment process and the customer relationship process through confirmation of order receipts and promised due dates.

- MPS also provides guidance to the sales group within the customer relationship process with respect to when future orders can be promised, and whether the due dates for existing orders can be adjusted in the time frame requested.

- The new service and product development process provides an updated bill of materials to the MRP system, and makes sure that every component and assembly needed for manufacturing of end items is properly recognized.

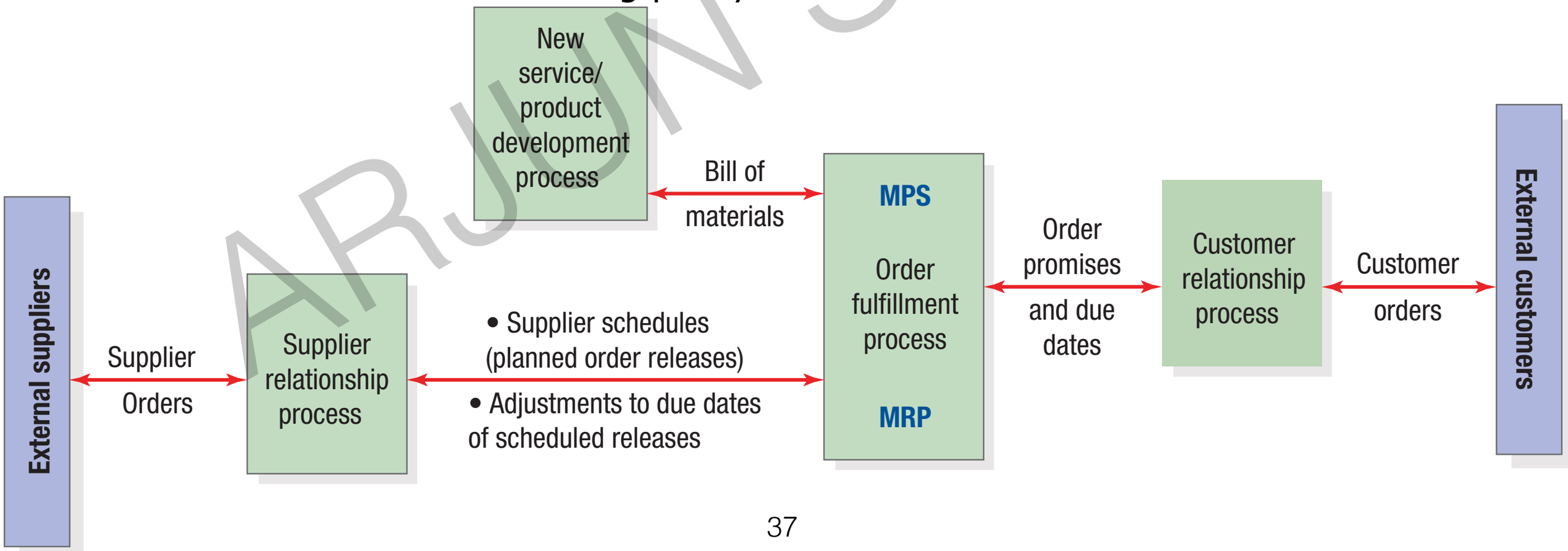
- In a similar vein on the inbound side, orders to external suppliers are based on the planned order releases, which come directly from the output of MRP reports.

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- The power of MRP, however, becomes evident when changes to an existing schedule are needed. These changes can be generated, for example, by changes to the MPS because a customer wants to change the timing or size of future orders, by some internal failure such as material shortages or unexpected machine downtime, or by supplier failure.

- In a supply chain, schedule changes have implications for customers as well as suppliers. Some firms, in partnership with their suppliers, have ERP/MRP systems that can actually "see" into their suppliers' inventory to determine if a particular item is in stock or, if not, when it can be expected.

- This is an advantage when contemplating a change to the original schedule of order releases. While systems such as this are powerful tools for making changes, care must be taken to avoid unnecessary fluctuations in the timing and size of PORs because of the choice of lot sizing policy.



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Resource Planning for Service Providers

- We have seen how manufacturing companies can disaggregate an MPS of finished products, which in turn must be translated into the needs for resources, such as staff, equipment, components, and financial assets. The driver for these resource requirements is a material requirements plan. Service providers, of course, must plan their resources just as manufacturers do. However, unlike finished goods, services cannot be inventoried. They must be provided on demand. In terms of resource planning then, service organizations must focus on maintaining the *capacity* to serve their customers. In this section, we will discuss how service providers use the concept of dependent demand and a bill of resources in managing capacity.

Dependent Demand for Services

- When we discussed planning and control systems for manufacturers earlier in this chapter, we introduced the concept of *dependent demand*, which is demand for an item that is a function of the production plans for some other item the company produces. For service resource planning, it is useful to define the concept of dependent demand to include demands for resources that are driven by forecasts of customer requests for services or by plans for various activities in support of the services the company provides.

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Bill of Resources

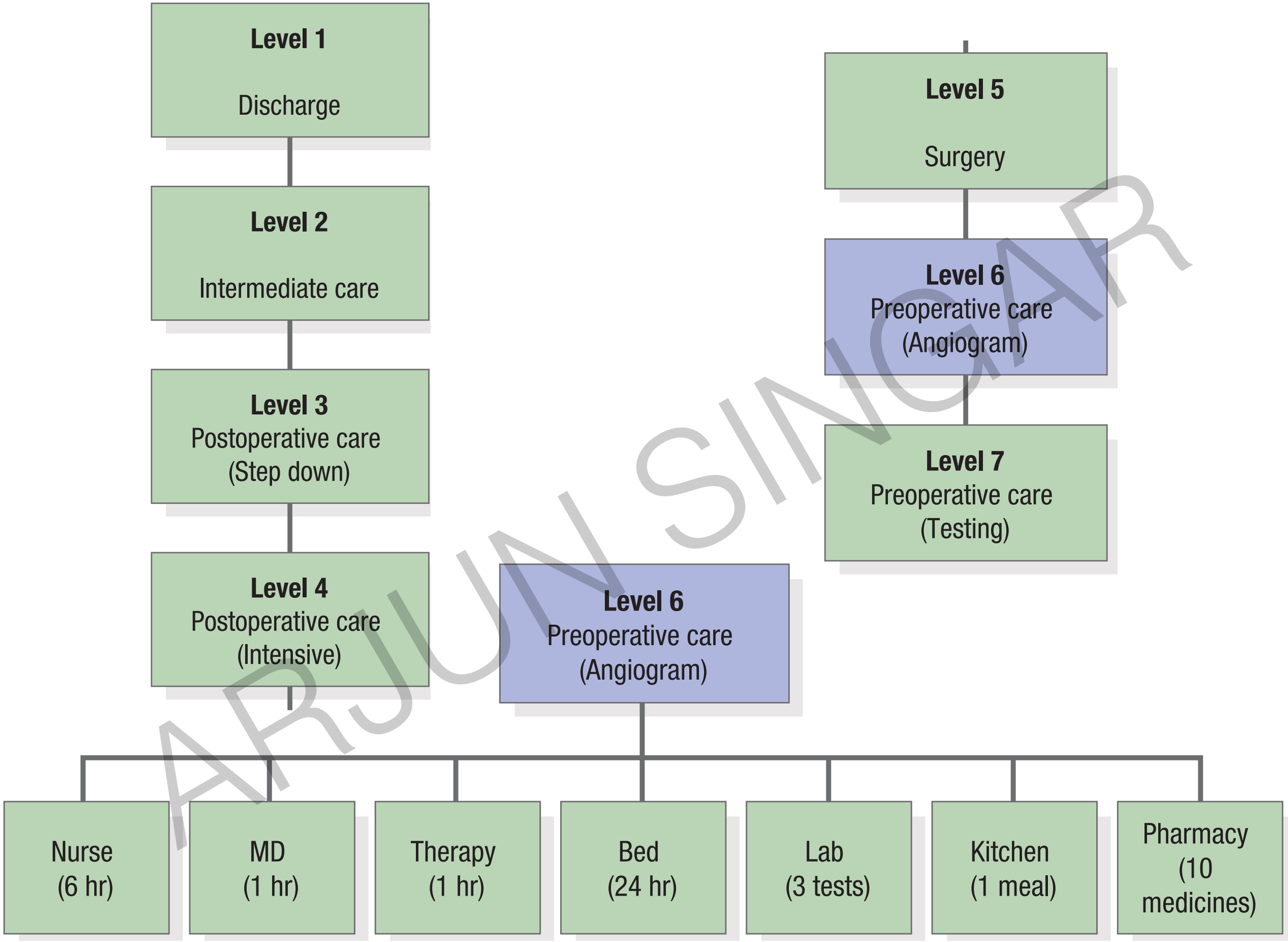
- The service analogy to the bill of materials in a manufacturing company is the **bill of resources (BOR)**, which is a record of a service firm's parent–component relationships and all of the materials, equipment time, staff, and other resources associated with them, including usage quantities.

- Once the service firm has completed its master schedule, the BOR can be used to determine what resources the firm will need, how much of them it will need, and when.

- A BOR for a service provider can be as complex as a BOM for a manufacturer. Consider a hospital that just scheduled treatment of a patient with an aneurysm. As shown in the following figure, the BOR for treatment of an aneurysm has seven levels, starting at the top (end item): (1) discharge, (2) intermediate care, (3) postoperative care (step down), (4) postoperative care (intensive), (5) surgery, (6) preoperative care (angiogram), and (7) preoperative care (testing).

- Each level of the BOR has a set of material and resource requirements and an associated lead time. For example, at level 6, shown in the following figure, the patient needs 6 hours of nurses' time, 1 hour of the primary MD's time, 1 hour of the respiratory therapist's time, 24 hours of bed time, 3 different lab tests, 1 meal, and 10 different medicines from the pharmacy. The lead time for this level is 1 day. The lead time for the entire stay for treatment of the aneurysm is 12.2 days. A master schedule of patient admissions and the BORs for each illness enable the hospital to manage its critical resources.

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